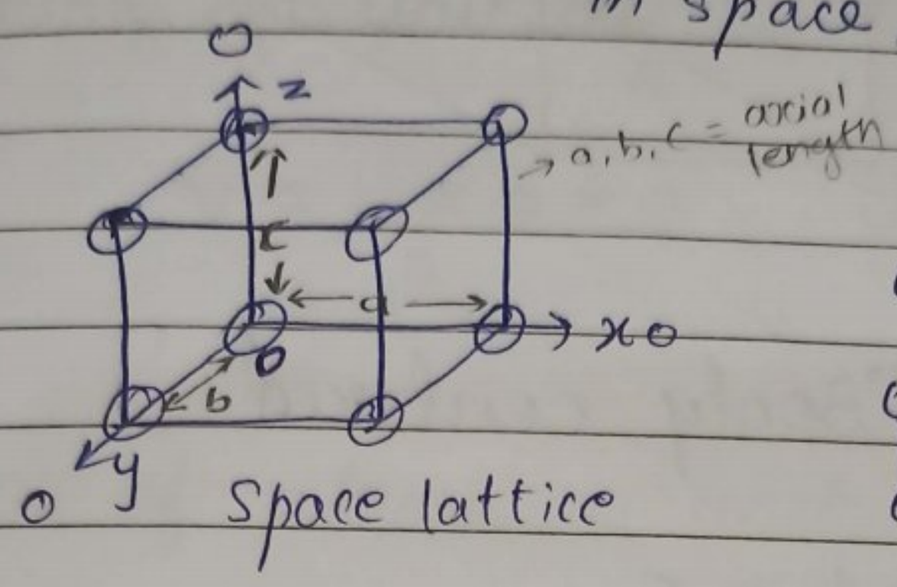
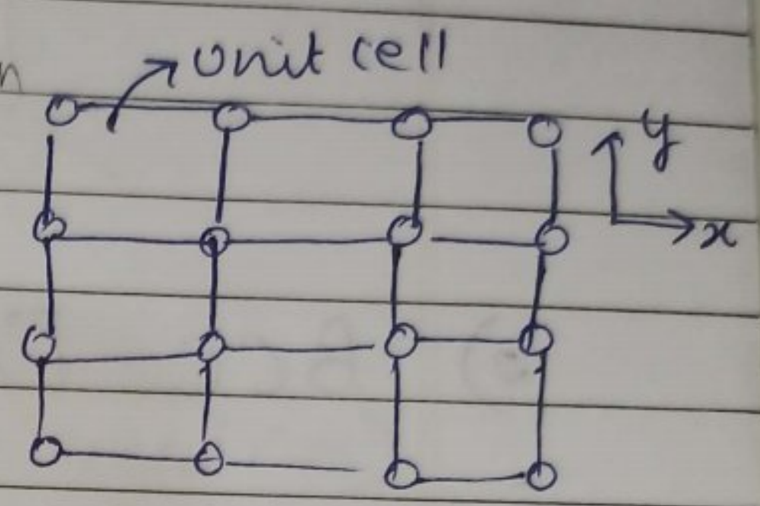
13/12/23

Crystal structure And X-Rays

space lattice = 3D arrangement of atoms in space.



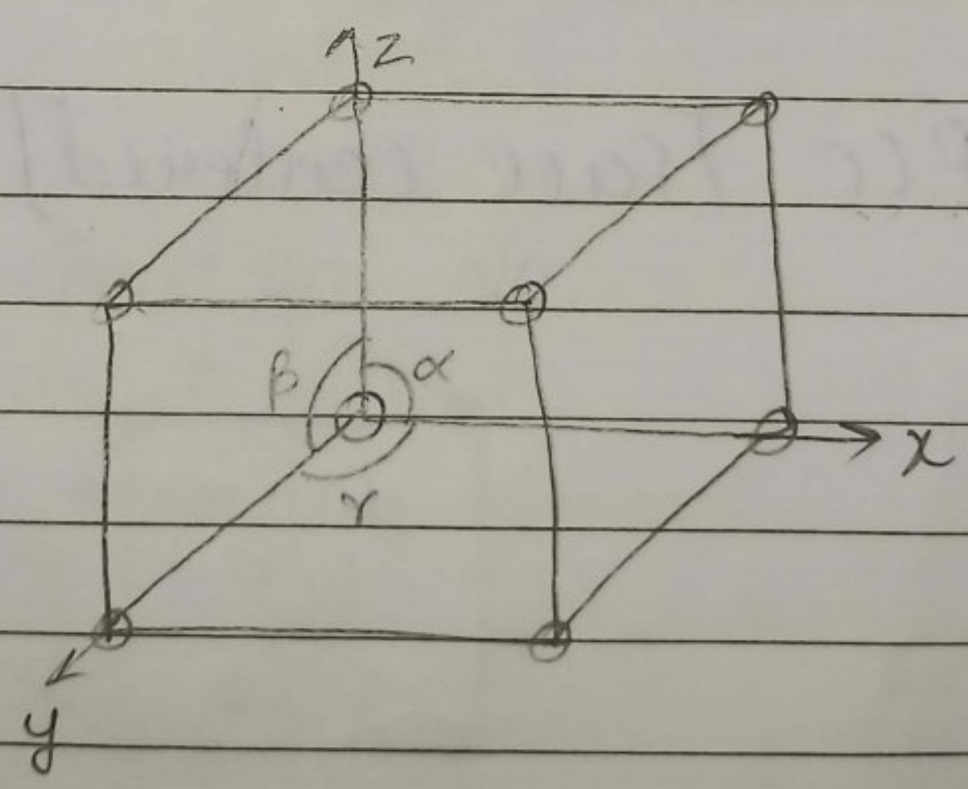
Space lattice



lattice point = no. of atoms arranged or combined together

Unit cell = ~~Small repeating of~~ smallest fundamental block of structure

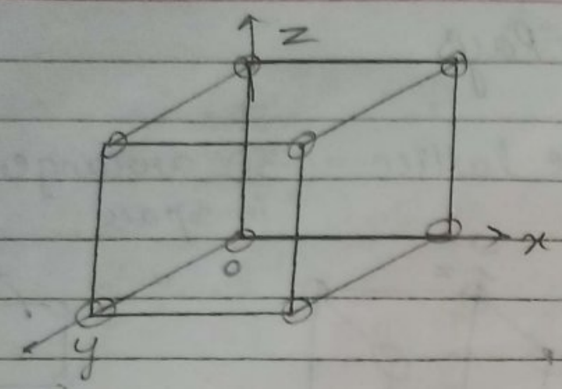
or small repeating structure



1) cubic [simple cubic cell (SCC)]

$a = b = c$

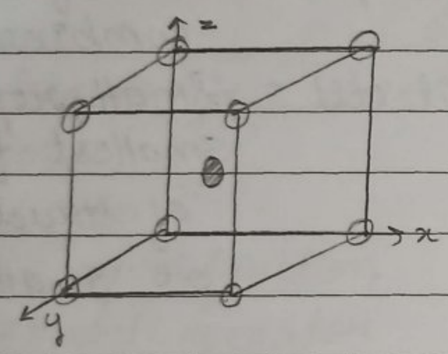
$\alpha = \beta = \gamma = 90°$



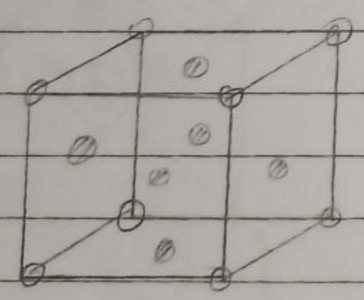
2) BCC [Body centered]

$a = b = c$

$\alpha = \beta = \gamma = 90^\circ$



3) FCC [Face centered]



$a = b = c$

$\alpha = \beta = \gamma = 90^\circ$

* Characteristics of unit cell

1. Volume of unit cell

$$V = a \times a \times a$$
$$= a^3$$

2. No's of atoms per unit cell (z)

$$SCC = \underbrace{\frac{1}{8}}_{\text{corner}} \times 8 = 1$$

$$BCC = \underbrace{\frac{1}{8}}_{\text{corner}} \times 8 + \underbrace{1}_{\text{Body}} = 2$$

$$FCC = \underbrace{\frac{1}{8}}_{\text{corner}} \times 8 + \underbrace{\frac{1}{2}}_{\text{face}} \times 6 = 4$$

3) Co-ordination No

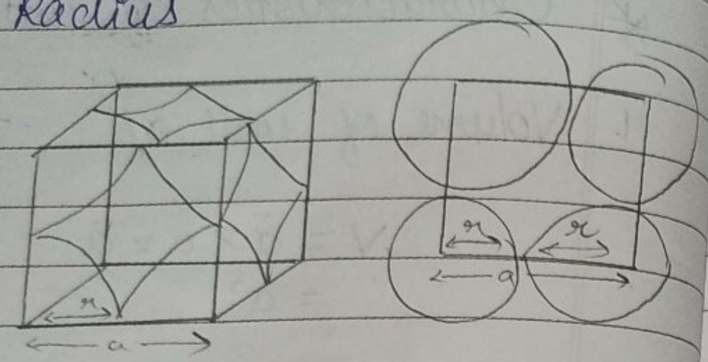
$SCC = 6 = CN$

$BCC = 8 = CN$

$FCC = 12 = CN$

4) Atomic Radius

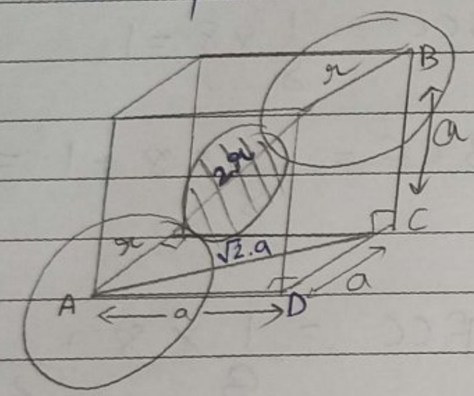
SCC =



$a = 2r$

$r = a/2$

BCC =

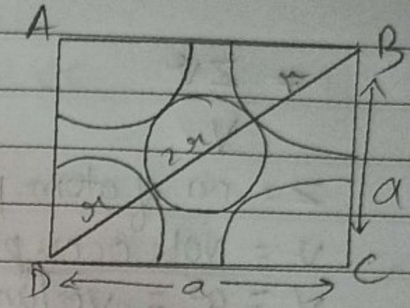


$$AC^2 = AD^2 + DC^2$$
$$= a^2 + a^2$$
$$= (\sqrt{2}a)^2$$
$$\boxed{AC = \sqrt{2}a}$$

In ΔACB

$$AB^2 = AC^2 + BC^2$$
$$(4r)^2 = (\sqrt{2}a)^2 + a^2$$
$$= 2a^2 + a^2$$
$$= 3a^2$$
$$= (\sqrt{3}a)^2$$
$$\boxed{r = \frac{\sqrt{3}a}{4}}$$

FCC =



In ΔBDC

$$BD^2 = DC^2 + BC^2$$
$$(4r)^2 = a^2 + a^2$$
$$= 2a^2$$
$$= (\sqrt{2}a)^2$$
$$4r = \sqrt{2}a$$
$$r = \frac{\sqrt{2}a}{4}$$
$$= \frac{\sqrt{2}a}{2\times 2}$$
$$= \frac{\sqrt{2}}{\sqrt{2}\times\sqrt{2}\times 2}a$$
$$\boxed{r = \frac{a}{2\sqrt{2}}}$$

** Atomic Packing Fraction [APF]

APF ~~is the~~ is ~~space~~ defined as the ratio of the volume occupied by the atoms contributed for the unit cell to the total volume of the unit cell.

$$APF = \frac{\text{Volume occupied by the atoms}}{\text{Total Volume of unit cell}}$$

Scanned with OKEN Scanner

$$= \frac{Zv}{V}$$

Z = no. of atom per unit cell

v = Vol. occupied by one atom

$V = a^3$ = volume of unit cell (cube)

- APF for SCC

$Z = 1$

$v = \frac{4}{3}\pi r^3$

$r = a/2 \qquad a = 2r$

$V = a^3$

$$APF = \frac{Zv}{V}$$

$$= \frac{1\left(\frac{4}{3}\pi r^3\right)}{a^3}$$

$$= \frac{1\left(\frac{4}{3}\pi (a/2)^3\right)}{}$$

$$= \frac{4\pi r^3}{3 \times (2r)^3}$$

$$= \frac{4\pi r^3}{3 \times 8 \times r^3}$$

$$= \frac{\pi}{6}$$

$$= 0.52$$

- APF for BCC

$Z = 2$

$v = \frac{4}{3}\pi r^3$

$r = \frac{\sqrt{3}\,a}{4} \qquad a = \frac{4r}{\sqrt{3}}$

$V = a^3$

$$APF = \frac{Zv}{V}$$

$$= \frac{2\left(\frac{4}{3}\pi r^3\right)}{a^3}$$

$$= \frac{8\pi r^3}{3\left(\frac{4r}{\sqrt{3}}\right)^3}$$

$$= \frac{8\pi r^3}{\cancel{3}8\left(\frac{64 r^3}{\cancel{3}\sqrt{3}}\right)}$$

$$= \frac{\sqrt{3}\pi}{8}$$

$$= 0.68$$

- APF for FCC

$Z = 4$

$v = 4/3\,\pi r^3$

$V = a^3$

$r = \frac{a}{2\sqrt{2}} \qquad a = 2\sqrt{2}\,r$

$$APF = \frac{Zv}{V}$$

$$= \frac{4\left(\frac{4}{3}\pi r^3\right)}{a^3}$$

$$= \frac{16\pi r^3}{3\,(2\sqrt{2}\,r)^3}$$

$$= \frac{16\pi r^3}{3 \times 8 \times 2\sqrt{2}\, r^3}$$

$$= \frac{2\pi}{3\sqrt{2}}$$

$$= 0.74$$

* Void space

$$\% \text{ void} = (1 - APF) \times 100\,\%$$

- $SCC = (1 - 0.52) \times 100$
  $= 48\,\%$

- $BCC = (1 - 0.68) \times 100$
  $= 32\%$

- $FCC = (1 - APF) \times 100$
  $= (1 - 0.74) \times 100$
  $= 26\%$

* Atomic Density (ρ)

$$\rho = \frac{m}{V} = \frac{\text{Total mass of the atom}}{\text{Total vol. of unit cell}}$$

$$= \frac{Zw}{V}$$

w = mass of one atom
V = volume of unit cell
Z = no. of atom per unit cell

$$V = a^3$$

$$\rho = \frac{Zw}{V}$$

$$\text{Mass of one atom} = \frac{\text{Atomic wt or molecular wt}}{\text{Avogadro Number}}$$

$$= \frac{M}{N_A}$$

$$N_A = 6.023 \times 10^{26} \text{ atom/kmole}$$

$$\therefore \rho = \frac{ZM}{N_A V}$$

$$\boxed{\rho = \frac{ZM}{N_A a^3}}$$

- SCC

$Z = 1$

$$\therefore \rho = \frac{1 \times M}{N_A \times a^3}$$

- BCC

$Z = 2$

$\therefore \rho = \frac{2M}{N_A a^3}$

- FCC

$Z = 4$

$\therefore \rho = \frac{4M}{N_A a^3}$

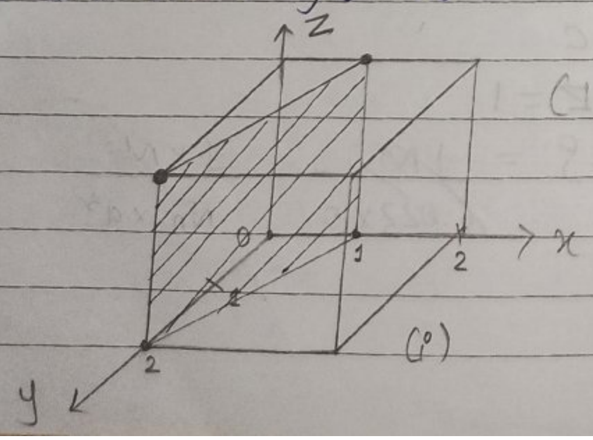
* Drawing of Planes

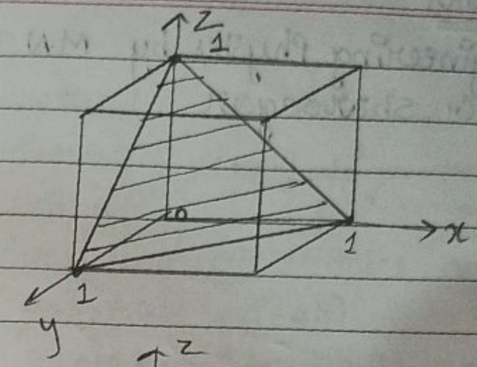
- Miller Indices :

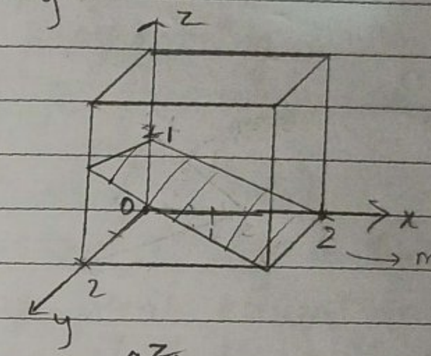
System of indices devised by Miller to study the arrangement of crystal planes in the unit cell ∴ It is called Miller Indices.

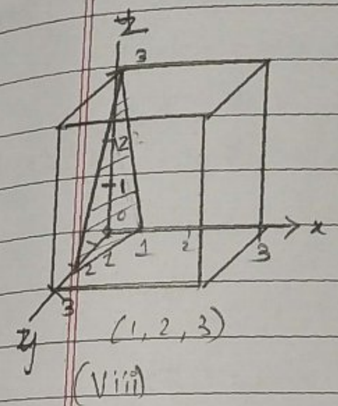
It is defined as the reciprocal of intercept made by the crystal plane along x, y, z axis when reduced to smallest set of integers.

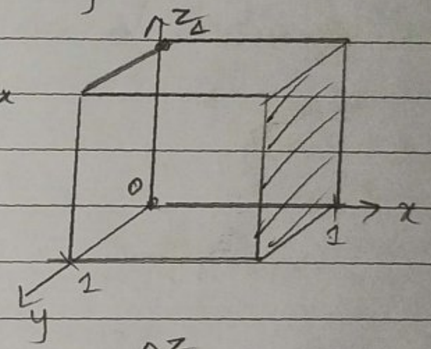
It is denoted by [hkl]

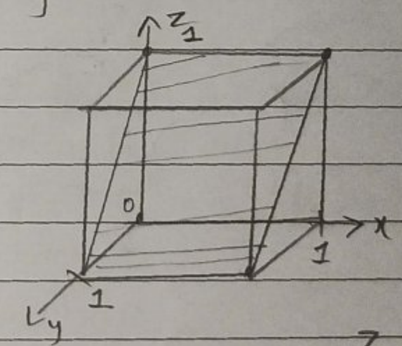
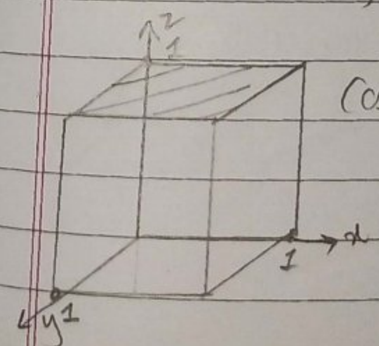
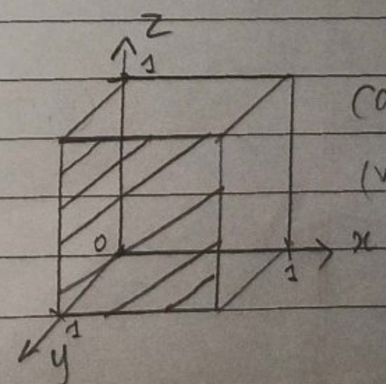












∞ means don't touch that axis